

Marks distribution for testing exercise 3			TOTAL
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3.33	3.33	3.33	10

1) Write at least the pseudocode corresponding to the identified method or methods .

(Code from the github repository)

2) Identify the test units.

Each method that has information relevant to the testing will be considered a test unit, so for this problem we would have:

Client.java:

- isAllowedToParticipate()

RecommendationService.java:

- isStayAtHome(), isSkiing(), isHikingOrClimbing(), isSummerAutumnSpringActivities(), isCulturalGastronomicActivities(), isBeachOrPoolActivities()

Venue.java:

- Venue(), withinCapacity()

Weather.java:

- Weather()

3) Identify the variables that must be considered to test the relevant methods included in the test units.

For isAllowedToParticipate() : the value to consider is a boolean (called perfectHealth)

For isStayAtHome(): two doubles(temperature and humidity) and a boolean(rainOrSnow)

For isSkiing(): two doubles (temperature and humidity) and two booleans (rainOrSnow and venueWithinCapapacity)

For isHikingOrClimbing(): one double(temperature) and two booleans(isRaining and venueWithinCapapacity)

For isSummerAutumnSpringActivities(): two doubles (temperature and humidity) and two booleans (isRaining and isCloudy)

For isCulturalGastronomicActivities(): a double (temperature) and two booleans (isRaining and venueWithinCapapacity)

For isBeachOrPoolActivities(): a double (temperature) and two booleans (isRaining and venueWithinCapapacity)

For Venue(): two integers(maxC and e)

For withinCapacity(): two integers(expectedAttendance and maxCapacity)

For Weather(): a double(humidity)

4) Identify the test values for each of the above variables, which will constitute the corresponding test cases using the three techniques discussed in theory (equivalence classes, boundary value analysis, and error guessing), specifying for each value obtained which technique has been used.

isAllowedToParticipate():

Variable	Partitions	Limit	Guess	Final	Explanation
perfectHealth	true		Yes	true,false	Some values are not included in the final column because they are being controlled in the command line interface, so they will never appear in our code.
	false		no		

isStayAtHome():

Variable	Partitions	Limit	Guess	Final	Explanation
temperature	-5.0	-0.1,0.0,0.1	“hello”	-5.0 , -0.1, 0.0, 0.1, 18.0	Some values are not included in the final column because they are being controlled in the command line interface, so they will never appear in our code.
	18.0		5,345(with commas)		

humidity	-5.0	-0.1,0.0,0.1	"hello"	0.0, 0.1, 10.0,14.99,15.0,15.1, 58.0
	58.0			
	10.0	14.99,15.0,15.1	43,67	true,false
rainOrSnow	true		1	
	false		0	

isSkiing():

Variable	Partitions	Limit	Guess	Final	Explanation
temperature	-7.0 32.48	-0.1,0.0,0.1	" _ "	18.0,- 7.0,32.48,0.0	Some values are not included in the final column because they are being controlled in the command line interface, so they will never appear in our code.
	18.0		WASD2		
humidity	10.0	14.0,15.0,16.0	tyu4	16.0,60.0,61.0- 12.0	
	62.0	59.0,60.0,61.0	4 6785		
	-12.0	-0.1,0.0,0.1	34,5		
rainOrSnow	true		1	true	
	false		0	false	
venueWithinCapacity	true		yes	true	
	false		no	false	

isHikingOrClimbing():

Variable	Partitions	Limit	Guess	Final	Explanation
temperature	-0.2	-0.1,0.0,0.1	3 5656	15.0,2.0,32.0,0.0,0.1	Some values are not included in the final column because they are being controlled in the command line interface, so they will never appear in our code.
	15.0				
	2.0	14.0,15.0,16.0	23,2		
	32.0				
isRaining	true		yes	true	
	false		no	false	
venueWithinCapacity	true		yes	true	
	false		no	false	

isSummerAutumnSpringActivities():

Variable	Partitions	Limit	Guess	Final	Explanation
temperature	-33.0 2.0	-0.1,0.0,0.1	"dsfs4"	32.0,25.0,2.0,14.0,-33.0	Some values are not included in the final column because they are being controlled in the command line interface, so they will never appear in our code.
	13.0 32.0	14.0,15.0,16.0	",,"		
	20.0	24.0,25.0,26.0			
humidity	10.0	14.0,15.0,16.0	tyu4	10.0, 58.0,61.0,90.0	
	58.0 90.0	59.0,60.0,61.0	4 6785 34.5		
	-1.0	-0.1,0.0,0.1			
isRaining	true		yes	true,false	
	false		no		
isCloudy	true		yes	true,false	
	false		no		

isCulturalGastronomicActivities():

Variable	Partitions	Limit	Guess	Final	Explanation
temperature	-50.0 2.0	-0.1,0.0,0.1	00,2	32.0,25.0,2.0,18.0,26.0,-50.0	Some values are not included in the final column because they are being controlled in the command line interface, so they will never appear in our code.
	18.0 27.0	24.0,25.0,26.0	"abc def ghi"		
	32.0	30.0,34.0,35.0,36.0	sdhf		
isRaining	true		yes	true,false	
	false		no		
venueWithinCapacity	true		yes	true,false	
	false		no		

isBeachOrPoolActivities():

Variable	Partitions	Limit	Guess	Final	Explanation
temperature	-0.7 28.0,18.0	-0.1,0.0,0.1	1,2,3,4	25.0,28.0,18.0,35.0,-0.7,-6.24	Some values are not included in the final column because they are being controlled in the command line interface, so they will never appear in our code.
	-6.24 25.0	30.0,34.0,35.0,36.0	“ ”		
isRaining	true		yes	true,false	
	false		no		
venueWithinCapacity	true		yes	true,false	
	false		no		

Venue():

Variable	Partitions	Limit	Guess	Final	Explanation
maxC	14,23,2000	-1,0,1	5 675	0,2000,23,14	Some values are not included in the final column because they are being controlled in the command line interface, so they will never appear in our code.
e	60,435,572,5726	-1,0,1	dg4	435,60,572,1,0	

withinCapacity():

Variable	Partitions	Limit	Guess	Final	Explanation
expectedAttendance	-23 2000	-1,0,1	5 675	2000,-23,1	Some values are not included in the final column because they are being controlled in the command line interface, so they will never appear in our code.
maximumCapacity	-1 60,5726	-1,0,1	dg4	60,5726, -1	

Weather():

Variable	Partitions	Limit	Guess	Final	Explanation
Humidity	-4.0 10.0,102.0	99.0,100.0,101.0	Hjkl,9	10.0, -4.0, 99.0,102.0,101.0	Some values are not included in the final column because they are being controlled in the command line interface, so they will never appear in our code.
	-4.0	-0.1,0.0,0.1	4 67852		

5) Calculate the maximum possible number of test cases that could be generated from the test values (combinatorial).

When combining all the possible inputs for every variable of each method, we would have:

isAllowedToParticipate() : 2

isStayAtHome(): $5 \times 7 \times 2 = 70$

isSkiing(): $4 \times 4 \times 2 \times 2 = 64$

isHikingOrClimbing(): $5 \times 2 \times 2 = 20$

isSummerAutumnSpringActivities(): $5 \times 4 \times 2 \times 2 = 80$

isCulturalGastronomicActivities(): $6 \times 2 \times 2 = 24$

isBeachOrPoolActivities(): $6 \times 2 \times 2 = 24$

Venue(): $4 \times 5 = 20$

withinCapacity(): $3 \times 3 = 9$

Weather(): 5

6) Define a set of test cases to fulfill each use (each value at least once).

To define this set cases, we will take into account the variable of each method that has the most values, which will be also the number of cases we will have for that method:

isAllowedToParticipate() :

Case	perfectHealth
1	true
2	false

isStayAtHome():

Case	temperature	humidity	rainOrSnow
1	-5.0	0.0	true
2	-0.1	0.1	false
3	0.0	10.0	true
4	0.1	14.99	false
5	18.0	15.0	true
6	0.0	15.1	true
7	-5.0	58.0	false

isSkiing():

Case	temperature	humidity	rainOrSnow	venueWithinCapacity
1	18.0	16.0	true	true
2	-7.0	60.0	false	true
3	32.48	61.0	true	false
4	0.0	-12.0	false	false

isHikingOrClimbing():

Case	temperature	isRaining	venueWithinCapacity
1	15.0	true	true
2	2.0	false	true
3	32.0	true	false
4	0.0	false	false
5	0.1	true	true

isSummerAutumnSpringActivities():

Case	temperature	humidity	isRaining	isCloudy
1	32.0	10.0	true	true
2	25.0	58.0	false	true
3	2.0	61.0	true	false
4	14.0	90.0	false	false
5	-33.0	58.0	true	false

isCulturalGastronomicActivities():

Case	temperature	isRaining	venueWithinCapacity
1	32.0	true	true
2	25.0	false	true
3	2.0	true	false
4	18.0	false	false
5	26.0	true	true
6	-50.0	false	false

isBeachOrPoolActivities():

Case	temperature	isRaining	venueWithinCapacity
1	28.0	true	true
2	25.0	false	true
3	35.0	true	false
4	18.0	false	false
5	-0.7	true	true
6	-6.24	false	false

Venue():

Case	maxC	e
1	0	0
2	2000	1
3	23	435
4	14	60
5	0	5726

withinCapacity():

Case	maximumCapacity	expectedAttendance
1	60	2000
2	5726	-23
3	-1	1

Weather():

Case	humidity
1	10.0
2	-4.0
3	99.0
4	102.0
5	101.0

7) Define test sets to achieve pairwise coverage using the algorithm explained in class. The results can be verified using the PICT program.

isAllowedToParticipate():

Case	perfectHealth
1	true
2	false

isStayAtHome():

Case	temperature	humidity	rainOrSnow
1	0,1	0,1	FALSE
2	-5	10	TRUE
3	-5	15,1	FALSE
4	18	58	FALSE
5	18	10	TRUE
6	-0,1	0	FALSE
7	0	15	FALSE
8	-5	14,99	TRUE
9	-5	0,1	TRUE
10	-5	58	TRUE
11	0	0	TRUE
12	0	0,1	TRUE
13	18	14,99	FALSE
14	-5	15	TRUE
15	-0,1	10	FALSE
16	-0,1	0,1	TRUE
17	0,1	0	TRUE
18	-0,1	14,99	FALSE
19	-5	0	TRUE
20	18	0,1	TRUE
21	18	15	TRUE
22	-0,1	15	FALSE
23	0,1	15	FALSE
24	0,1	58	TRUE
25	0	15,1	TRUE
26	0	10	FALSE

27	18	0	FALSE
28	0,1	15,1	TRUE
29	0	58	TRUE
30	0,1	14,99	FALSE
31	-0,1	58	TRUE
32	0,1	10	FALSE
34	0	14,99	TRUE
35	-0,1	15,1	TRUE
36	18	15,1	FALSE

isSkiing():

Case	temperature	humidity	rainOrSnow	venueWithinCapacity
1	0.0	16.0	TRUE	TRUE
2	-7.0	16.0	FALSE	FALSE
3	-7.0	-12.0	TRUE	TRUE
4	18.0	-12.0	FALSE	FALSE
5	0.0	61.0	FALSE	FALSE
6	-7.0	60.0	FALSE	TRUE
7	18.0	60.0	TRUE	FALSE
8	18.0	16.0	TRUE	TRUE
9	32.48	60.0	TRUE	FALSE
10	32.48	-12.0	FALSE	TRUE
11	0.0	60.0	FALSE	FALSE
12	18.0	61.0	TRUE	TRUE
13	-7.0	61.0	TRUE	FALSE
14	32.48	16.0	TRUE	TRUE
15	0.0	-12.0	TRUE	TRUE
16	32.48	61.0	FALSE	FALSE

isHikingOrClimbing():

Case	temperature	isRaining	venueWithinCapacity
1	2.0	TRUE	FALSE
2	15.0	FALSE	FALSE
3	32.0	FALSE	TRUE
4	0.1	TRUE	TRUE
5	32.0	TRUE	FALSE
6	0.0	FALSE	FALSE
7	2.0	FALSE	TRUE
8	0.0	TRUE	TRUE
9	15.0	TRUE	TRUE
10	0.1	FALSE	FALSE

isSummerAutumnSpringActivities():

Case	temperature	humidity	isRaining	isCloudy
1	25.0	58.0	TRUE	TRUE
2	-33.0	58.0	FALSE	FALSE
3	-33.0	61.0	TRUE	TRUE
4	32.0	58.0	FALSE	TRUE
5	32.0	90.0	TRUE	FALSE
6	25.0	10.0	FALSE	FALSE
7	2.0	61.0	FALSE	FALSE
8	14.0	90.0	FALSE	TRUE
9	14.0	61.0	TRUE	FALSE
10	32.0	10.0	TRUE	TRUE
11	25.0	90.0	FALSE	TRUE
12	32.0	61.0	TRUE	TRUE

13	25.0	61.0	FALSE	TRUE
14	-33.0	90.0	TRUE	TRUE
15	14.0	10.0	FALSE	FALSE
16	-33.0	10.0	FALSE	TRUE
17	2.0	10.0	TRUE	TRUE
18	14.0	58.0	FALSE	TRUE
19	2.0	58.0	FALSE	FALSE
20	2.0	90.0	FALSE	TRUE

isCulturalGastronomicActivities():

Case	temperature	isRaining	venueWithinCapacity
1	26.0	TRUE	FALSE
2	25.0	TRUE	TRUE
3	18.0	TRUE	TRUE
4	25.0	FALSE	FALSE
5	32.0	FALSE	TRUE
6	2.0	FALSE	TRUE
7	26.0	FALSE	TRUE
8	2.0	TRUE	FALSE
9	32.0	TRUE	FALSE
10	-50.0	FALSE	FALSE
11	18.0	FALSE	FALSE
12	-50.0	TRUE	TRUE

isBeachOrPoolActivities():

Case	temperature	isRaining	venueWithinCapacity
1	-0.7	TRUE	FALSE
2	28.0	TRUE	TRUE
3	35.0	TRUE	TRUE
4	28.0	FALSE	FALSE
5	25.0	FALSE	TRUE
6	18.0	FALSE	TRUE
7	-0.7	FALSE	TRUE
8	18.0	TRUE	FALSE
9	25.0	TRUE	FALSE
10	-6.24	FALSE	FALSE
11	35.0	FALSE	FALSE
12	-6.24	TRUE	TRUE

Venue():

Case	maxC	e
1	14	435
2	0	0
3	23	5726
4	0	435
5	23	435
6	23	1
7	14	1
8	23	60
9	2000	0
10	0	60

11	14	60
12	14	5726
13	0	5726
14	2000	1
15	14	0
16	2000	60
17	2000	435
18	2000	5726
19	0	1
20	23	0

withinCapacity():

Case	expectedAttendance	maximumCapacity
1	-23	60
2	-23	-1
3	2000	5726
4	1	60
5	2000	-1
6	1	-1
7	-23	5726
8	2000	60
9	1	5726

Weather():

Case	humidity
1	102.0
2	99.0
3	10.0
4	101.0
5	-4.0

8) For code sections involving decisions, propose a set of test cases to achieve decision coverage.

Decision coverage happens when a decision is evaluated to true and false at least once. Taking this into account:

Test ID	Method	perfectHealth	infectiousDiseaseLast2Weeks	temperature	humidity	isRaining	isSnowing	isCloudy	maxCapacity	expectedVisitors	Expected Result
D1	isAllowedToParticipate	true	false	-	-	-	-	-	-	-	true
D2	isAllowedToParticipate	false	false	-	-	-	-	-	-	-	false
D3	isStayAtHome	true	false	-5.0	10.0	true	false	-	-	-	STAY-HOME
D4	isStayAtHome	true	false	18.0	58.0	false	false	-	-	-	false
D5	isSkiing	true	false	-7.0	16.0	false	false	-	50	40	Skiing
D6	isSkiing	true	false	32.48	61.0	true	false	-	50	40	false
D7	isHikingOrClimbing	true	false	0.1	-	false	false	-	30	20	Hiking Or Climbing
D8	isHikingOrClimbing	true	false	32.0	-	true	false	-	30	20	false
D9	isSummerAutumnSpringActivities	true	false	20.0	58.0	false	false	false	-	-	Spring, Summer and Autumn Activities

D10	isSummerAutumnSpringActivities	true	false	32.0	90.0	true	false	true	-	-	false
D11	isCulturalGastronomicActivities	true	false	27.0	-	false	false	-	100	80	Cultural or Gastronomic activities
D12	isCulturalGastronomicActivities	true	false	26.0	-	true	false	-	100	80	false
D13	isBeachOrPoolActivities	true	false	35.0	-	false	false	-	200	150	Beach or Pool activities
D14	isBeachOrPoolActivities	true	false	-6.24	-	true	false	-	200	150	false
D15	withinCapacity	-	-	-	-	-	-	-	60	50	true
D16	withinCapacity	-	-	-	-	-	-	-	60	2000	false
D17	Weather	-	-	-	10.0	-	-	-	-	-	valid
D18	Weather	-	-	-	102.0	-	-	-	-	-	invalid

9) For code sections involving decisions, propose a set of test cases to achieve MC/DC coverage.

MC/DC coverage is met when every possible value of a conditions determines the output of the decision at least once.

isAllowedToParticipate():

Test	perfectHealth	infectiousDiseaseLast2Weeks	Result
A1	true	false	true
A2	false	false	false
A3	true	true	false

isStayAtHome():

Test	temperature	humidity	rainOrSnow	temp<0	hum<15	rainOrSnow	Result
S1	-5.0	10.0	true	T	T	T	true
S2	18.0	10.0	true	F	T	T	false
S3	-5.0	58.0	true	T	F	T	false
S4	-5.0	10.0	false	T	T	F	false

isSkiing():

Test	temperature	humidity	rainOrSnow	venueWithinCapacity	temp<0	hum<15	!rainOrSnow	venueOK	Result
K1	-7.0	16.0	false	true	T	F	T	T	false
K2	-7.0	10.0	false	true	T	T	T	T	true
K3	-7.0	10.0	true	true	T	T	F	T	false
K4	-7.0	10.0	false	false	T	T	T	F	false

isHikingOrClimbing():

Test	temperature	isRaining	venueWithinCapacity	tempOK	!isRaining	venueOK	Result
H1	0.1	false	true	T	T	T	true
H2	18.0	false	true	F	T	T	false
H3	10.0	true	true	T	F	T	false
H4	10.0	false	false	T	T	F	false

isSummerAutumnSpringActivities():

Test	temperature	humidity	isRaining	isCloudy	tempOK	!isRaining	!isCloudy	humOK	Result
SA1	20.0	58.0	false	false	T	T	T	T	true
SA2	32.0	58.0	false	false	F	T	T	T	false
SA3	20.0	70.0	false	false	T	T	T	F	false
SA4	20.0	58.0	true	false	T	F	T	T	false
SA5	20.0	58.0	false	true	T	T	F	T	false

isCulturalGastronomicActivities():

Test	temperature	isRaining	venueWithinCapacity	tempOK	!isRaining	venueOK	Result
CG1	27.0	false	true	T	T	T	true
CG2	18.0	false	true	F	T	T	false
CG3	27.0	true	true	T	F	T	false
CG4	27.0	false	false	T	T	F	false

isBeachOrPoolActivities():

Test	temperature	isRaining	venueWithinCapacity	tempOK	!isRaining	venueOK	Result
BP1	35.0	false	true	T	T	T	true
BP2	28.0	false	true	F	T	T	false
BP3	35.0	true	true	T	F	T	false
BP4	35.0	false	false	T	T	F	false

withinCapacity():

Test	expectedVisitors	maxCapacity	Result
V1	50	60	true
V2	2000	60	false

10) Comment on the results regarding the number of test cases obtained in sections 4, 5, and 6: what can be said about the coverage achieved? To what extent does the implementation of the program influence the design and implementation of the test cases??

The selected test cases are significantly fewer than the full combinatorial set, but are sufficient to ensure each input value is exercised at least once, which is the goal of value coverage. The logical structure of the program directly influences the number and design of tests required for MC/DC and decision coverage.

Because the max combinatorial test set is impractical to run in full, the reduced sets in section 6 are efficient, still ensuring that each value is used at least once, each condition in the code is exercised for decision coverage and each condition can independently affect the outcome(MC/DC). The proper use of equivalence classes, boundary analysis and MC/DC allows for efficient but effective testing.

ClientTest:

isAllowedToParticipate():

Input value	Expected result
(true,false)	True
(false,false)	False
(true,true)	False
(false, true)	false

IsInPerfectHealth ():

Input value	Expected result
(true,false)	True
(false,false)	False

testHasInfectuousDeseaseLast2Weeks()

Input value	Expected result
(true,true)	True
(true,false)	False

RecommendationTest**RecommendActivityClientNotAllowed()**

Input value	Expected result
(false, true, 40.0, 80.0, true, false, false, 50, 100)	“NO ACTIVITY”
(true, true, 45.0, 90.0, true, true, true, 100, 200)	“NO ACTIVITY”

IsStayAtHome()

Input value	Expected result
(0.1, 0.1, false, false, false)	false
(-5.0, 10.0, true, false, false)	true

(-5.0, 15.1, false, false, false)	False
(18.0, 58.0, false, false, false)	False
(-0.1, 0.0, false, false, false)	false
(0.0, 15.0, false, false, false)	false
(-5.0, 14.99, true, false, false)	true
(-5.0, 0.1, true, false, false)	true
(-5.0, 58.0, true, false, false)	false
(0.0, 0.0, true, false, false)	false
(0.0, 0.1, true, false, false)	false
(18.0, 14.99, false, false, false)	false
(-5.0, 15.0, true, false, false)	False
(-0.1, 10.0, false, false, false)	False
(-0.1, 0.1, true, false, false)	true
(0.1, 0.0, true, false, false)	False
(-0.1, 14.99, false, false, false)	false
(-5.0, 0.0, true, false, false)	True
(18.0, 0.1, true, false, false)	False
(18.0, 15.0, true, false, false)	false
(-0.1, 15.0, false, false, false)	False
(0.1, 15.0, false, false, false)	false
(0.1, 58.0, true, false, false)	False
(0.0, 15.1, true, false, false)	false
(0.0, 10.0, false, false, false)	False
(18.0, 0.0, false, false, false)	false
(0.1, 15.1, true, false, false)	False
(0.0, 58.0, true, false, false)	false
(0.1, 14.99, false, false, false)	False
(-0.1, 58.0, true, false, false)	false
(0.1, 10.0, false, false, false)	False

(0.0, 14.99, true, false, false)	False
(-0.1, 15.1, true, false, false)	False
(18.0, 15.1, false, false, false)	False

IsSkiing()

Input value	Expected result
(-5.0, 10.0, false, false, false)	True
(0.0, 16.0, true, false, false)	False
(-7.0, 16.0, false, false, false)	False
(-7.0, -12.0, true, false, false)	False
(18.0, -12.0, false, false, false)	False
(0.0, 61.0, false, false, false)	False
(-7.0, 60.0, false, false, false)	False
(18.0, 60.0, true, false, false)	False
(18.0, 16.0, true, false, false)	False
(32.48, 60.0, true, false, false)	False
(32.48, -12.0, false, false, false)	False
(0.0, 60.0, false, false, false)	False
(18.0, 61.0, true, false, false)	False
(-7.0, 61.0, true, false, false)	False
(32.48, 16.0, true, false, false)	False
(0.0, -12.0, true, false, false)	false
(32.48, 61.0, false, false, false)	false

IsHikingOrClimbing()

Input value	Expected result
(2.0, 50.0, true, false, false)	False
(15.0, 50.0, false, false, false)	False
(32.0, 50.0, false, false, false)	False
(0.1, 50.0, true, false, false)	False
(32.0, 50.0, true, false, false)	False
(0.0, 50.0, false, false, false)	False
(2.0, 50.0, false, false, false)	true
(0.0, 50.0, true, false, false)	False
(15.0, 50.0, true, false, false)	False
(0.1, 50.0, false, false, false)	false

IsSummerAutumnSpringActivities()

Input value	Expected result
(20.0, 58.0, false, false, false)	True
(25.0, 58.0, true, false, true)	false
(-33.0, 58.0, false, false, false)	False
(-33.0, 61.0, true, false, true)	False
(32.0, 58.0, false, false, true)	False
(32.0, 90.0, true, false, false)	False
(25.0, 10.0, false, false, false)	False
(2.0, 61.0, false, false, false)	False
(14.0, 90.0, false, false, true)	False
(14.0, 61.0, true, false, false)	False
(32.0, 10.0, true, false, true)	False
(25.0, 90.0, false, false, true)	False

(32.0, 61.0, true, false, true)	False
(25.0, 61.0, false, false, true)	False
(-33.0, 90.0, true, false, true)	False
(14.0, 10.0, false, false, false)	False
(-33.0, 10.0, false, false, true)	False
(2.0, 10.0, true, false, true)	false
(14.0, 58.0, false, false, true)	False
(2.0, 58.0, false, false, false)	False
(2.0, 90.0, false, false, true)	false

IsCulturalGastronomicActivities()

Input value	Expected result
(26.0, 50.0, true, false, false)	False
(25.0, 50.0, true, false, false)	False
(18.0, 50.0, true, false, false)	False
(25.0, 50.0, false, false, false)	False
(32.0, 50.0, false, false, false)	False
(2.0, 50.0, false, false, false)	False
(26.0, 50.0, false, false, false)	True
(2.0, 50.0, true, false, false)	False
(32.0, 50.0, true, false, false)	False
(-50.0, 50.0, false, false, false)	False
(18.0, 50.0, false, false, false)	False
(-50.0, 50.0, true, false, false)	false

IsBeachOrPoolActivities()

Input value	Expected result
(32.0, 50.0, false, false, false)	True
(-0.7, 50.0, true, false, false)	False
(28.0, 50.0, true, false, false)	False
(35.0, 50.0, true, false, false)	False
(28.0, 50.0, false, false, false)	False
(25.0, 50.0, false, false, false)	False
(18.0, 50.0, false, false, false)	False
(-0.7, 50.0, false, false, false)	False
(18.0, 50.0, true, false, false)	False
(25.0, 50.0, true, false, false)	False
(-6.24, 50.0, false, false, false)	False
(35.0, 50.0, false, false, false)	False
(-6.24, 50.0, true, false, false)	false

NoActivityMatch()

Input value	Expected result
Weather(40.0, 80.0, true, false, false); Venue(50, 100)	True
Weather(45.0, 90.0, true, true, true) Venue(100, 200)	True

Venue

WithinCapacity()

Input value	Expected result
(5726,2000)	True
(60, 1)	True
(60, 2000)	False
(5726, 1)	true